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Key Points:

- We built a machine learning model to automatically find fast forward interplanetary shocks using Wind satellite data
- Against the 185-event composite benchmark, our method achieves 82% recall with an 11% false alarm rate
- Rankine-Hugniot checks confirm 67 additional shocks beyond the Center for Astrophysics/Helsinki list, including 11 events independently listed in Oliveira (2023, https://doi.org/10.3389/fspas.2023.1240323)

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Automatic Identification of Interplanetary Shocks Based on Machine Learning

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Abstract Fast forward interplanetary shocks (FFs) are one of the primary drivers of space weather events. This study presents an automated detection algorithm for FFs based on a multilayer perceptron model, utilizing in situ measurements of interplanetary magnetic fields and solar wind plasma from the Wind spacecraft at 1 AU. The training data set was constructed using FF events from the Harvard-Smithsonian Center for Astrophysics (CfA) shock list between 1995 and 2019, while the testing data set comprised FF events from a combined catalog of the CfA and Helsinki University shock lists from 2020 to 2024. During the testing period, the method identified 169 FFs, of which 83 matched entries in the existing catalog. Among the remaining 86 uncatalogued events, 67 satisfied the Rankine-Hugoniot (R-H) jump conditions, confirming them as novel shocks not previously recorded. Using direct comparison with the composite CfA/Helsinki catalog, the model achieves a recall of 80.58%. Since some apparent false positives in this direct comparison are later supported as physically consistent shocks by the R-H checks, the combined evaluation against the expanded benchmark yields a recall of 81.62% with false alarm rate = 10.65%. This approach provides a robust tool for the classification and study of interplanetary shock properties, thus enhancing the capability for space weather forecasting.

Plain Language Summary Interplanetary shocks are solar perturbations that can disturb Earth's magnetic environment and affect satellites, power grids, and communications. We used measurements from NASA's Wind spacecraft to develop an algorithm to recognize fast forward (FF) shocks directly in the data stream. Unlike earlier techniques that rely on hand-set thresholds, our model learns the characteristic patterns of a shock from multiple measurements at once. After training on past, well-studied events, we tested the method on Wind data obtained in recent years. It found 169 shocks, of which 150 were confirmed as genuine FF shocks, including 67 that were missing from existing shock lists but satisfied standard physics criteria. Overall, the method recovered about 81.62% of real shocks while keeping false alarms near 10.65%, clearly improving on previous approaches. This automated, accurate detection will help researchers build better shock catalogs and improve space-weather forecasting.

1. Introduction

Interplanetary (IP) shocks, particularly fast forward (FF) shocks, are one of the primary drivers of significant space weather phenomena. They are often driven by coronal mass ejections (CMEs) or stream interaction regions (Cane & Richardson, 2003; Chi et al., 2016, 2018; Jian et al., 2006; Richardson & Cane, 2010). Upon reaching Earth, these shocks can energize particles and compress the magnetosphere, producing sudden geomagnetic disturbances. When accompanied by sustained southward interplanetary magnetic field (IMF) conditions $(B_z < 0)$, they can contribute to the onset and intensification of geomagnetic storms and associated geomagnetic variations that can drive geomagnetically induced currents (e.g., Cash et al., 2014; Oliveira et al., 2024; Wu & Lepping, 2016). These effects collectively pose significant risks to human daily life and space-based assets (Baker et al., 2018; Cane et al., 1988; Carter et al., 2015; Desai & Giacalone, 2016; McPherron, 1979; Oliveira & Ngwira, 2017; Pulkkinen et al., 2017; Reames, 1999; Richardson & Cane, 2011).The properties of IP shocks have therefore long been a major research focus in heliophysics and space weather studies (Kilpua et al., 2015; Trotta et al., 2025). Consequently, the rapid and accurate identification of IP shocks is critical in the domain of space weather forecasting.

IP shocks are characterized by abrupt jumps in solar wind speed, density, temperature, and IMF strength observed in situ. Some widely used shock catalogs have relied, at least in part, on visual/manual inspection of solar-wind and IMF time series (Lumme et al., 2024). To improve the efficiency of shock identification, automatic approaches have also been developed that applied threshold criteria to upstream and downstream parameter variations (Neugebauer et al., 2003; Vorotnikov et al., 2008). These methods were subsequently refined by combining multiple thresholds of solar wind and IMF parameters to enhance both efficiency and accuracy (Cash et al., 2014; Kruparova et al., 2013; Lumme et al., 2024; Trotta et al., 2025). However, threshold-based techniques have notable limitations. Overly strict thresholds may exclude weak but physically valid shocks, whereas overly lenient ones often generate false positives due to solar wind fluctuations or small-scale transient structures such as other types of discontinuities including itermediate shocks (Tsurutani et al., 2011). Moreover, these approaches typically rely on moving-average calculations of upstream and downstream parameters, which inevitably reduce the temporal resolution of the data and smooth out abrupt variations. As a result, the sharp and transient features that are intrinsic to IP shocks are often obscured, particularly in high-cadence observations.

With the advancement of artificial intelligence, machine learning (ML) approaches have been progressively integrated into space weather research, demonstrating significant potential in various domains such as solar wind categorization and prediction of solar eruptions, solar energetic particles, solar wind speed, CME arrival time, the occurrence rates of shocks, geomagnetic activities (Camporeale et al., 2018; H. Li et al., 2020; Y. Li et al., 2024; R. Lin et al., 2024; J. Liu et al., 2018; E. Liu et al., 2020; Oliveira et al., 2024; Roy et al., 2025; Siciliano et al., 2021; Upendran et al., 2020; Wang et al., 2023; Yang & Shen, 2019; Yang et al., 2018). Unlike classical detection schemes that rely on fixed jump thresholds or moving-average windows, ML algorithms can ingest multi-parameter and automatically extract nonlinear combinations of features indicative of shock ramps. This capability enables the detection of shocks that might be smoothed out or misclassified by conventional methods. A major challenge for supervised ML is that existing shock catalogs are valuable but not exhaustive benchmarks. As a result, different catalogs compiled from the same solar wind observations can show only partial overlap because of differences in event selection criteria and quality-control procedures. Similar issues have been noted in ML-based studies of interplanetary CME detection (Chen et al., 2022; Nguyen et al., 2019, 2025; Rüdisser et al., 2022). Recently, ML-based techniques have been employed to identify IP shock candidates (Lumme et al., 2024). The method is implemented as InterPlanetary Support Vector Machine (IPSVM) and provides candidate events from solar wind measurements. Nevertheless, confirming these candidates as IP shocks still relies on additional threshold checks using solar wind plasma and IMF data.

In this study, we propose an automatic identification method for IP shocks based on a supervised machine-learning model. Unlike conventional methods, our approach eliminates the reliance on predefined thresholds for jump conditions, thus significantly improving the effectiveness of IP shock detection. In Section 2, we provide a detailed description of the data and the method, including data preparation, model architecture, output post-processing, and model performance evaluation. Section 3 presents the identification results obtained with our model. Finally, a summary of the key theoretical contributions and potential future research directions is provided in Section 4.

2. Data and Method

2.1. Data and Shock Lists

This method uses five key parameters: magnetic field intensity, solar wind speed, proton density, temperature, and pressure. The solar wind pressure is calculated as the sum of magnetic pressure and thermal pressure, with the assumption of $T_e = T_p$, where T_e and T_p denote electron and proton temperatures, respectively. In this work, 3-s resolution IMF measurements and solar wind plasma properties (bulk speed, density, temperature) obtained from Wind spacecraft's Magnetic Field Investigation (MFI, Lepping et al., 1995) and Three-Dimensional Plasma (3DP, R. P. Lin et al., 1995) instruments were used for model training, validation, and testing. To be consistent with catalogs' shock analyses, the "k0" level plasma data from Wind spacecraft's Solar Wind Experiment (SWE, Ogilvie et al., 1995) were used for Rankine-Hugoniot (R-H) verification.

We utilized shock events observed by the Wind spacecraft, which were obtained from two IP shock databases: the IP shock database from the Harvard-Smithsonian CfA (https://lweb.cfa.harvard.edu/shocks, hereinafter referred to as CfA catalog) and the University of Helsinki (https://ipshocks.helsinki.fi, hereinafter referred to as Helsinki catalog). Considering that the shock events in CfA catalog have been further confirmed by Mach numbers and R-H relations for shocks, the FF events during 1995–2019 in the CfA catalog were employed to create a training data set for multilayer perceptron (MLP), while the FF events during 2020–2024 in the combined CfA and Helsinki catalogs were used for testing. To build the composite benchmark catalog, we merged the CfA and Helsinki lists and removed duplicates by time matching (within 1 min), so each physical shock is counted once. The combined list for the testing period contains 103 FF shock events.

Because the reference catalogs are not expected to be exhaustive, we do not anticipate perfect agreement in a direct catalog-to-catalog comparison. In particular, a non-negligible fraction of apparent false positives may correspond to genuine shocks that are absent from the chosen reference list. We also use the interplanetary shock list compiled by Oliveira (2023) as an independent cross-catalog reference. The catalog is constructed from L1 observations (Wind/ACE/DSCOVR) and reports geomagnetic-impact diagnostics (e.g., the minimum SMR within 2 hours after the listed time), which we use only for supplementary consistency checks rather than for defining our primary benchmark.

2.2. Method

The flowchart of this automated IP shock detection method consists of three sequential components: data preparation, model processing, and post-processing to output the time of the detected shock ramp.

2.2.1. Data Preparation

The data are pre-processed before input into the MLP model. The preprocessing pipeline includes four steps: temporal alignment, data cleaning, pressure calculation, and data normalization.

To maintain temporal consistency, IMF intensity data were linearly interpolated onto the 3DP instrument's temporal grid, ensuring alignment with solar wind plasma parameters. Due to occasional errors in data timing, data cleaning is required by merging temporally duplicated adjacent records and filling in missing data points for MFI and 3DP data. Moreover, if a long time interval between adjacent data points was encountered, additional entries were inserted. Their timestamps were determined by linear interpolation, while all other parameters were set to null.

We generate samples using a sliding window of 20 data points with a step size of 1 data point, producing overlapping 20-point windows. For each 20-point sample window, we form a 60-point normalization window by extending 20 data points to both the left and right. The normalization (min–max scaling to [0, 1]) is computed from this 60-point window and applied to the same interval. The normalized central 20 points are then used as the input sample, as illustrated in Figure 1. Although neighboring 20-point samples partially overlap in time, they are not identical. Moreover, each sample is normalized using its own 60-point local background window. Therefore overlapping windows generally yield different normalized inputs and do not introduce duplicate samples across labels or splits.

To mitigate the impact of irregular solar wind disturbances on model performance, we defined four distinct sample categories: FF, FF upstream, FF downstream, and non-FF samples. To create the samples for model training and validation, we first determine the shock ramp boundaries ($t_{rampstart}$, $t_{rampend}$) for each event by case-by-case manual inspection. To ensure that the 60-point normalization window always contains the whole ramp and a portion of the upstream/downstream plasma, samples are labeled based on their temporal location relative to ($t_{rampstart}$, $t_{rampend}$): samples whose 20-point window fully falls within ($t_{rampend} - 20\Delta t$, $t_{rampstart} + 20\Delta t$) are labeled as "FF"; samples within ($t_{rampend} - 40\Delta t$, $t_{rampstart}$) are labeled as "FF upstream" ("UP" for short); and samples within ($t_{rampend}$, $t_{rampstart} + 40\Delta t$) are labeled as "FF downstream" ("DOWN" for short), where Δt is the data cadence. Samples whose 20-point window partially overlaps the ramp interval does not satisfy any of the above FF/UP/DOWN ranges and are discarded (not assigned to any class). "Non-FF" samples are drawn from days without cataloged shocks, selecting at most one segment per day, ensuring no overlap among non-FF samples and avoiding leakage from temporally adjacent windows. A schematic illustration of these four cases is provided in Figure 1. Because FF samples are defined to include the full ramp plus a small margin of upstream/downstream points, a limited temporal overlap with neighboring UP/DOWN samples can occur. However, this overlap is restricted to non-ramp background points adjacent to the ramp, and any sample that intersects the ramp only partially is discarded and receives no label, and each 20-point sample is assigned to at most one class by construction. No interpolation is applied to data gaps. Samples containing NaNs are excluded from the training and validation sets. During testing, such samples are assigned an additional "invalid" label, not as a physical class

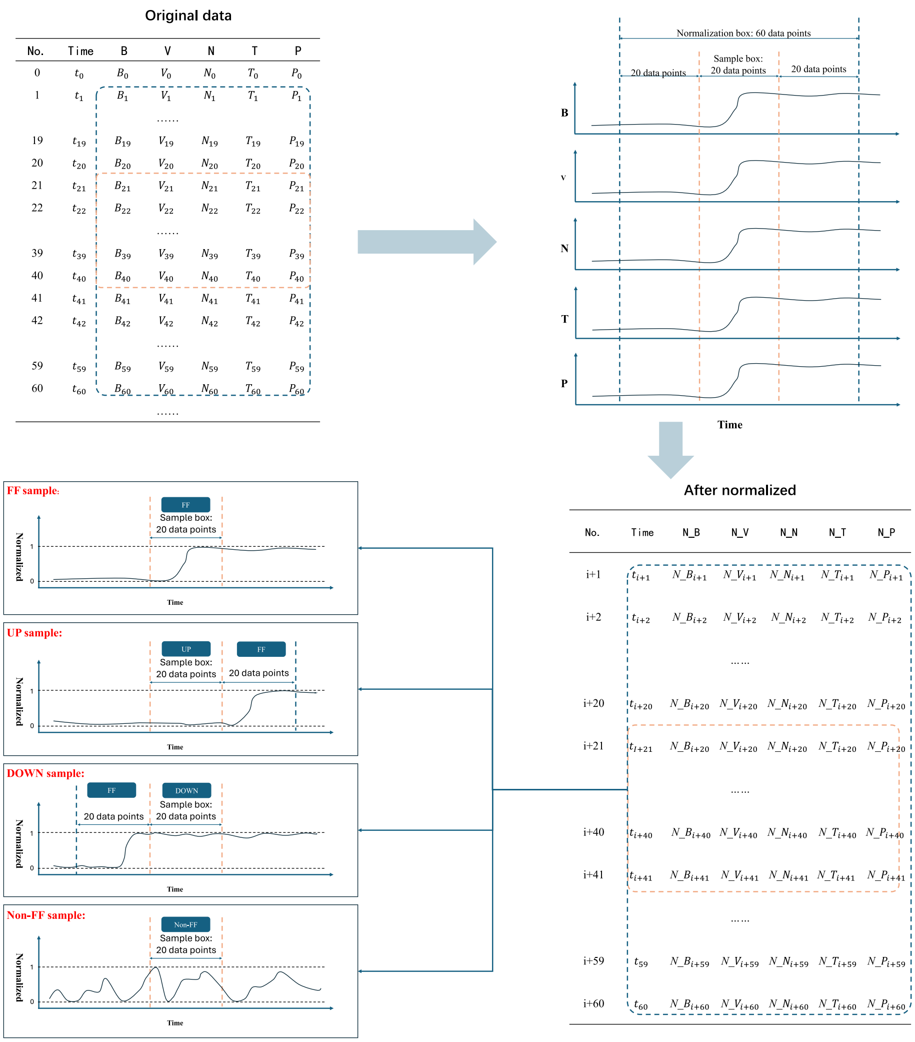

Figure 1. Schematic of sample construction and labeling. A sliding window is extracted from the original time series and normalized over a larger context window. The central "sample box" is then used as the model input. Examples illustrate how samples are assigned to the four classes (fast forward [FF], UP, DOWN, and non-FF) based on the relative position of the shock ramp within the sample box (see Section 2.2.1 for details).

for shock identification, but as a marker of missing-data samples. This preserves the temporal continuity of the prediction sequence during post-processing, since removing these samples entirely could lead to missed shock events.

The quantitative composition and distribution of the training data set are summarized in Table 1. Samples of the first three categories (FF, FF upstream, and FF downstream) were constructed by combining FF events from the CfA catalog (1995–2019) with observational data from the Wind spacecraft. After removing entries containing null values, this yielded 5,111 FF samples, 5,280 FF upstream samples, and 5,290 FF downstream samples. Non-FF samples were generated by randomly selecting normalization windows from periods between 1995 and 2019 that were devoid of interplanetary shock events. This set was then randomly subsampled to match the size of the other three categories and further cleaned by removing null values, resulting in 5,326 non-FF samples to ensure a balanced data set.

2.2.2. The Multilayer Perceptron Model

The MLP, a fundamental neural network model in ML, comprises an input layer, one or more hidden layers for feature abstraction, and an output layer for generating results. In this study, we employed the MLP for its proven

Table 1
Quantitative Composition and Distribution of the Training Data Set

Sample type	Number	Percentage (%)
FF upstream samples	5,280	25.13
FF samples	5,111	24.33
FF downstream samples	5,290	25.18
Non-FF samples	5,326	25.35

capability in modeling complex, non-linear relationships within high-dimensional data, which is essential for accurately identifying shock signatures from solar wind parameters. Its critical components include activation functions, loss functions, and optimizers: the activation functions are non-linear units in the hidden layers that endow the MLP model with nonlinear fitting capacity; the loss function measures the model performance by calculating the discrepancy between true and predicted values, guiding the optimizer in weight parameter updates; during each training iteration, the optimizer adjusts the network weights based on the loss function, enabling the MLP to effectively capture characteristics of the underlying data (Gardner & Dorling, 1998; Popescu et al., 2009).

In this work, we used the rectified linear unit as the activation function to improve computational efficiency. The cross-entropy loss, which measures the mismatch between predicted class probabilities and the true labels, was used to train the classifier. We used stochastic gradient descent with a fixed learning rate of 0.01 and did not employ any learning-rate scheduler. The detailed structure of the MLP is presented in Table 2.

2.2.3. Post-Processing

In this method, the spacecraft data were processed into discrete, independent samples that were fed into the model, resulting in a sequence of individual label outputs as conceptually illustrated in Figure 2. However, relying solely on these discrete classifications is insufficient for robust FF event identification, due to the continuous nature of shock structures in the solar wind. Moreover, transient solar wind perturbations may produce output sequences that sporadically resemble FF features. Therefore, post-processing that analyzes the temporal continuity and contextual consistency of the output label sequence is essential to ensure accurate identification of FF shocks and return the time of shock ramps.

The post-processing method is established through the following criteria. Three fundamental concepts are introduced: FF group, upstream group, downstream group, and null. An FF group refers to a consecutive sequence of outputs classified as FF samples. The upstream group comprises the 20 output samples immediately preceding the current FF group, while the downstream group consists of the 20 output samples directly following the FF group. And the null means the value of output is null because of data gaps in input data. Then, five conditional criteria are subsequently applied: (a) whether the FF group length exceeds 10 but remains below 20 samples; (b) whether the value of final output in the upstream group is null and the predominant output within the upstream group is null; (c) whether the value of initial output in the downstream group is null and the predominant output within the downstream group is null; (d) whether the predominant output type within the upstream group (excluding null values) corresponds to FF upstream samples; (e) whether the predominant output type in the downstream group (excluding null values) matches FF downstream samples.

Table 2
Detailed Structure of the Multilayer Perceptron (MLP) Used in This Work

Task type	Four-class classification + invalid output	
Model	Structures	Parameters
MLP	First hidden layer	Linear (100, 64)
	Second hidden layer	Linear (64, 32)
	Output layer	Linear (32, 5)
	Activation function	ReLU (hidden layers)
	Loss function	Cross-entropy loss
	Optimizer	SGD

Note. The network outputs five logits: four correspond to the physical classes used for training/validation, while the additional invalid output is used only during testing to flag NaN-containing samples. These samples are excluded from training and validation and are not treated as a physical class for shock identification.

These criteria collectively were used to ensure an FF event, as illustrated in Figure 3. After that, the central output within the FF group of an FF is selected to find the time of the shock ramp. Among the 20 data points corresponding to this output, the average of the time values of the two adjacent data points with the largest jump of input parameters is taken as the time of the shock ramp representing the FF event. Note that this jump-based operation is used only to assign a representative ramp time after an FF event has been identified by the MLP outputs and the post-processing criteria.

2.3. Performance Metrics

To evaluate the event-level detection performance of the algorithm, we compare the model-identified shock ramp times with entries in the reference catalog. Under this definition, true positive (TP) represents the number of instances that are actually FF event and are correctly detected as FF by the model; FP represents the number of targets that are actually non-FF but are incorrectly identified as FF by the model; false negative (FN) represents the number of samples that are actually FF but are incorrectly classified as non-

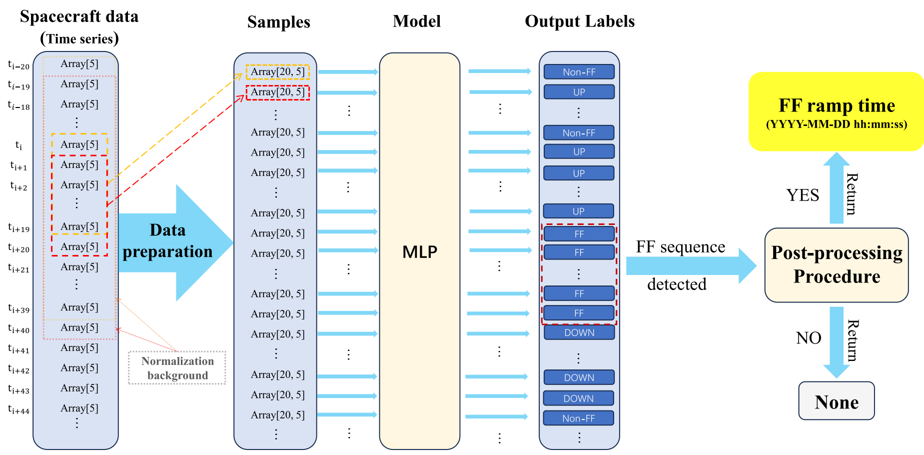

Figure 2. Diagram of the process from spacecraft data to final outputs. Here "UP" refers to "fast forward [FF] upstream" samples and "DOWN" refers to "FF downstream" samples.

FF by the model, also known as the number of misses. In practice, we define a TP when a model-identified shock ramp time matches an entry in the reference catalog within ±1 min, otherwise the detection is counted as a FP. Likewise, any catalog entry without a matched model detection within ±1 min is counted as a FN. We adopt ±1 min because shock ramps typically last seconds to tens of seconds, and the CfA catalog reports times at minute-level precision whereas the Helsinki catalog is at second-level precision, so a ±1 min tolerance is a practical and physically meaningful matching criterion.

Based on the above TP/FP/FN definitions, five evaluation metrics were used to assess its FF identification capability: Threat Score (TS), FAR, Precision, Recall, and F1-score. The TS, also termed the Critical Success Index, quantifies prediction accuracy and is widely applied in meteorology and disaster forecasting, with its formulation given in Equation 1. The FAR, representing erroneous target detection under null conditions, is calculated via Equation 2. Precision, Recall, and F1-score, as common evaluation metrics in ML, are defined in Equations 3–5 (Cash et al., 2014; Goutte & Gaussier, 2005). Note that with the definitions in Equations 2 and 3, FAR = 1-Precision. Therefore, we report FAR and omit Precision in the tables to avoid redundancy.

$$TS = \frac{TP}{TP + FP + FN} \tag{1}$$

$$FAR = \frac{FP}{TP + FP} \tag{2}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{3}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{4}$$

$$F1\text{-score} = \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \tag{5}$$

To enable a direct and transparent comparison of detection efficiency with a representative threshold-based approach, we implemented the method of Cash et al. (2014) following the algorithm described in Figure 3 of that paper. We then evaluated both algorithms using the same Wind test data set and the same reference shock list, and applied the identical event-matching criterion (±1 min) to define TP/FP/FN for a consistent comparison.

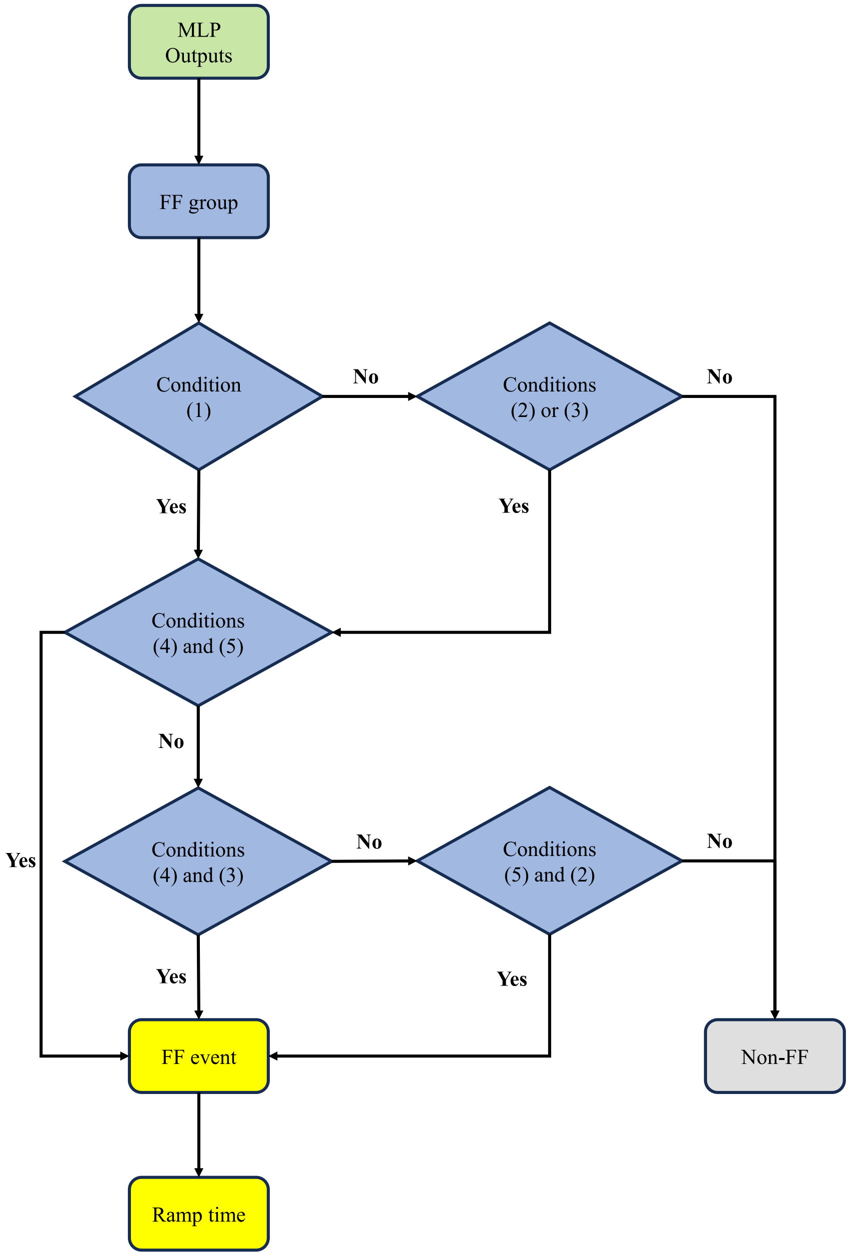

Figure 3. Diagram of the post-processing from multilayer perceptron model outputs (individual labels) to final outputs (fast forward [FF] events represented by ramp time or non-FF).

3. Results

3.1. Model Training and Validation

The training data set was partitioned into training and validation sets at a 7:3 ratio, and the loss curve documented the evolution of the loss function value across training epochs, serving as a critical indicator for assessing training completion. The model underwent 1,000 training epochs to ensure convergence and derive optimal weight parameters. As evidenced by the loss curve in Figure 4, a rapid decline in loss values occurs during initial epochs. Subsequent iterations exhibit loss stabilization, indicating model convergence. This convergence behavior confirms the model's capacity to capture underlying data patterns.

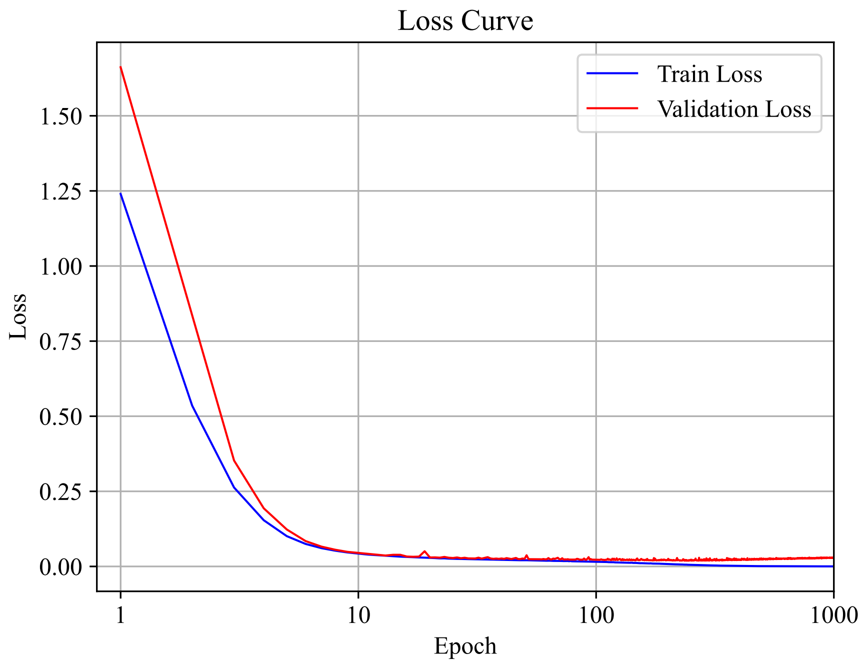

Figure 4. Loss curve during the training epochs.

The Receiver Operating Characteristic (ROC) curve and the Area Under the Curve (AUC) are standard metrics for evaluating classification models (Hanley & McNeil, 1982). The ROC curve plots the False Positive Rate (FPR) against the TP Rate (TPR) across varying thresholds, illustrating the trade-off between sensitivity and specificity. Here TPR = TP/(TP + FN) and FPR = FP/(FP + TN), both defined on individual labeled samples in the validation set. The AUC quantifies the overall performance of the model, with values ranging from 0.5 (indicating no discrimination ability) to 1.0 (indicating perfect discrimination). These ROC/AUC results evaluate pointwise classification on the validation set, whereas the event-level detection metrics in Section 2.3 are computed for the testing period after post-processing.

During the 1,000 training epochs, the weight parameters corresponding to the epoch with minimal validation loss were selected as optimal. To comprehensively assess the model's classifying capability, the ROC curves were generated for each category, accompanied by corresponding AUC values. As shown in Figure 5, the model demonstrates robust recognition capabilities for all sample categories in the validation set.

3.2. Initial Model Performance: Direct Comparison With the Combined Catalog

This model was applied to IMF and solar wind plasma data acquired by Wind spacecraft from 2020 to May 2024 for testing purpose. During the testing period, a total of 169 FF events were identified. Among them, 83 matched entries in the composite CfA and Helsinki catalogs (total 103 FFs), corresponding to a recall of 80.58%. Table 3 summarizes performance metrics computed by direct comparison with the composite CfA and Helsinki catalog, and compares our results with a reproduced implementation of a threshold-based approach following Cash et al. (2014). Both algorithms were run with the same Wind test interval and evaluated against the same composite CfA–Helsinki shock list. Here, the TP (N_{hit}) is the number of the targeted results that are in the combined Cfa and Helsinki catalog.

Compared with the reproduced implementation following Cash et al. (2014), our method achieves a higher TS and F1-score, and a lower FAR, while maintaining a comparable recall. At the same time, the FAR remains relatively high, and the TS is still moderate, due to the 86 FPs. Since the CfA and Helsinki catalogs mutually miss FF events, this outcome may be primarily attributed to the fact that the CfA and Helsinki catalogs do not encompass all FF events observed by the Wind spacecraft during the testing period. This incompleteness is also supported by an independent comparison with the Oliveira shock list (Oliveira, 2023). Among the events that are not included in either the CfA or Helsinki catalog, we find that 11 are present in Oliveira list, suggesting that at least a subset of the apparent "false positives" in the direct catalog-to-catalog comparison may correspond to genuine shocks.

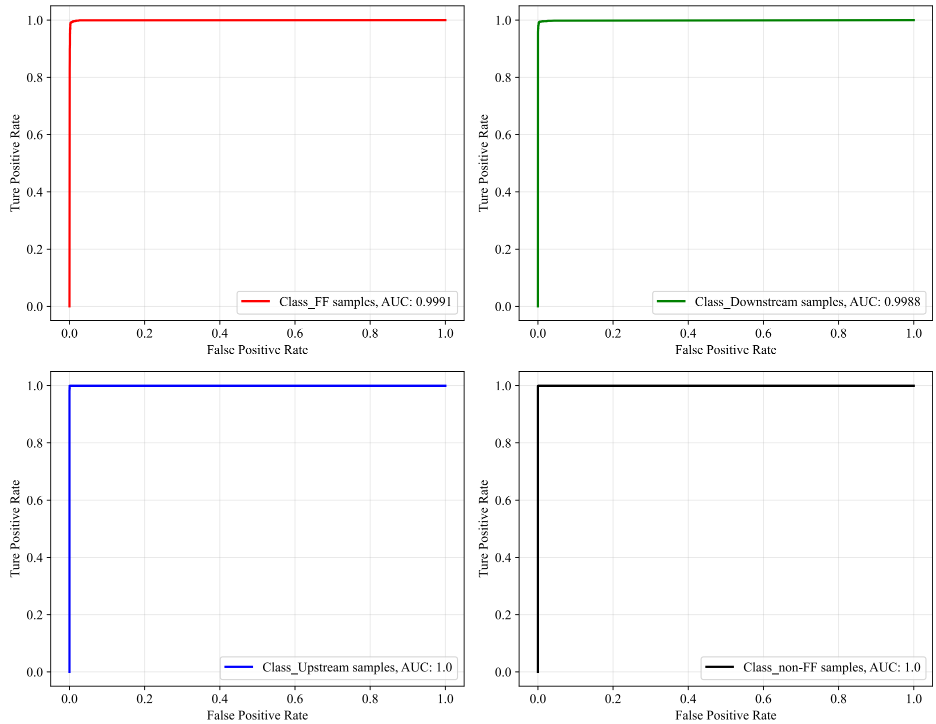

Figure 5. Receiver operating characteristic curve for sample categorizing.

Consequently, some genuine FFs identified by this model are absent from these catalogs, leading to their misclassification as FP in this direct comparison. To address this limitation, we conducted an R-H verification on the 86 FF events identified by this model that are not listed in either catalog, aiming to validate their authenticity as genuine FFs.

3.3. Rankine-Hugoniot Relationship Analysis of Uncatalogued FFs

The following conditions were applied as the criteria of fast IP shocks: (a) The upstream fast Mach number ($M_f = v/v_{\text{fast}}$, where v denotes the flow speed in the shock frame and v_{fast} is the fast magnetosonic speed) is greater than 1, and the downstream fast Mach number is less than 1; (b) The upstream and downstream parameters satisfy the Rankine-Hugoniot (R-H) relations (Burlaga, 1995). We use the root mean square RH_{overall} of the following five parameters to examine the R-H relations, which are the ratios of the left-hand side to the right-hand side of the five R-H equations:

$$RH_1 = \frac{\rho_2 v_{2n}}{\rho_1 v_{1n}} - 1 \tag{6}$$

$$RH_2 = \frac{\rho_2 v_{2n}^2 + P_2 + \frac{B_2^2}{8\pi}}{\rho_1 v_{1n}^2 + P_1 + \frac{B_1^2}{8\pi}} - 1 \tag{7}$$

$$RH_3 = \frac{\rho_2 v_{2n} v_{2t} + \frac{B_n B_{2t}}{4\pi}}{\rho_1 v_{1n} v_{1t} + \frac{B_n B_{1t}}{4\pi}} - 1 \tag{8}$$

$$RH_4 = \frac{\left(\frac{\rho_2 v_2^2}{2} + \frac{5P_2}{2} + \frac{B_{2t}^2}{4\pi}\right) v_{2n} - \frac{B_n B_{2t} v_{2t}}{4\pi}}{\left(\frac{\rho_1 v_1^2}{2} + \frac{5P_1}{2} + \frac{B_{1t}^2}{4\pi}\right) v_{1n} - \frac{B_n B_{1t} v_{1t}}{4\pi}} - 1 \tag{9}$$

$$RH_5 = \frac{B_{2n}}{B_{1n}} - 1 \tag{10}$$

Table 3
Model Performance (Evaluated by Direct Comparison) of Our Approach and a Reproduced Threshold-Based Method Following Cash et al. (2014)

	This method	Cash et al. (2014)
Reference shocks	103	103
Hits (TP)	83	**89**
Misses (FN)	20	**14**
False positives (FP)	**86**	170
Threat score (TS)	**0.4392**	0.3260
False alarm rate (FAR)	**0.5089**	0.6564
Recall	0.8058	**0.8641**
F1-score	**0.6103**	0.4917

Note. The bold values indicate the best result of each metric.

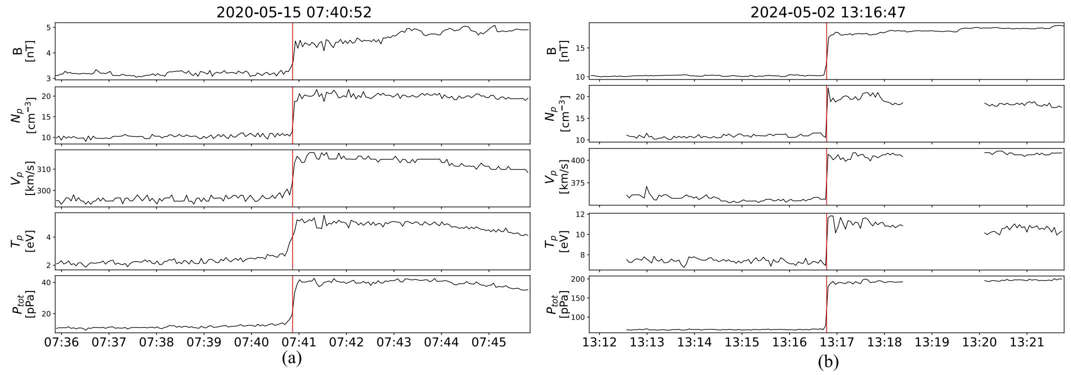

Figure 6. Two uncatalogued fast forward (FF) events that meet the physical analysis criteria. The red vertical lines indicate the FF shock ramps with time information returned from this method. (a) shock event identified by our model on May 15, 2020, which is not included in Cfa-Helsinki catalog; (b) another shock event identified by our model on May 02, 2024, also not included in Cfa-Helsinki catalog.

$$RH_{\mathrm{overall}} = \sqrt{\frac{RH_1^2 + RH_2^2 + RH_3^2 + RH_4^2 + RH_5^2}{5}} \tag{11}$$

Here, subscript "1" and subscript "2" denote upstream and downstream averages, while subscript "n" and subscript "t" denote normal and tangential components of the corresponding parameters, respectively. ρ is mass density, v is plasma bulk flow speed in the shock frame, P is the total (sum of thermal and magnetic) pressure, B is the magnetic field.

Because R-H evaluation in in situ data depends on the shock normal estimate and the upstream/downstream averaging windows, R-H relations are inherently window-dependent and exact equality is not expected. We therefore scan 5-min windows within ±10 min on each side of the ramp and report the best-fit RH_{overall} values. Events with best-fit $RH_{\mathrm{overall}} < 0.1$ are regarded as consistent with R-H relations.

3.4. True Model Performance: Combined Comparison With Catalogs and R-H Analysis

The results of the R-H analysis revealed that 67 out of the 86 uncatalogued FFs satisfy the two physical criteria, indicating these FFs are novel shocks absent from the present catalogs. Two examples of the uncatalogued FF events are shown in Figure 6. For a consistent benchmark, we applied the same R-H verification procedure to the discrepant candidates detected by the reproduced threshold-based method of Cash et al. (2014). This analysis validates 37 shocks, of which 22 overlap with our newly identified events, while 15 are unique to the Cash-based detections. Accordingly, we constructed an augmented reference list consisting of the 103 CfA-Helsinki catalog shocks, plus 67 RH—validated shocks identified by our model, and 15 R-H validated shocks unique to the Cash-based method, yielding 185 reference shocks in total. Using this augmented list as the benchmark catalog, Table 4 reports the final performance metrics for both methods.

The performance of this model to identify FFs is significantly superior to previous methods, as evidenced by the comparative metrics presented in Table 4. The high TS (0.7438) and Recall (81.62%) values indicate that this model can effectively identify FF events, while the low FAR (10.65%) values demonstrate that the majority of FFs identified by this model are genuine shocks. The F1-score (85.31%) further confirms the model's robust performance in FF identification. The ability to detect 67 previously uncatalogued FF events further underscores the model's effectiveness compared to prior manual or threshold-based approaches.

Despite the high performance of this model, there are still some limitations. We manually analyzed the reason for the 20 undetected FFs, as summarized in Table 5. For a subset of these cases, the discontinuity is not visually prominent in the 3DP plasma data (especially temperature, see Figure 7a), which may reduce the model's confidence in identifying an FF group. In the majority of cases, data gaps occur at or across the shock ramp, as shown in Figure 7b. Because no interpolation is applied and samples containing NaNs are treated as invalid in our implementation, the method cannot provide a valid detection when key ramp measurements are missing.

Table 4
True Model Performance (Evaluated by Combined Catalog Comparison and R-H Verification) of Our Approach and a Reproduced Threshold-Based Method Following Cash et al. (2014)

	This method	Cash et al. (2014)
Reference shocks	185	185
Hits (TP)	**151**	125
Misses (FN)	**34**	60
False positives (FP)	**18**	134
Threat score (TS)	**0.7438**	0.3919
False alarm rate (FAR)	**0.1065**	0.5174
Recall	**0.8162**	0.6757
F1-score	**0.8531**	0.5631

Note. The bold values indicate the best result of each metric.

4. Summary

In this paper, we present a novel machine-learning-based framework for the automatic identification of IP shocks, with a particular focus on FF shocks that play a key role in driving space weather phenomena. Unlike traditional threshold-based methods, our approach employs a MLP model trained on high-resolution (3-s) in situ measurements from the Wind spacecraft, including IMF strength, solar wind speed, proton density, temperature, and the calculated total pressures. By eliminating subjective threshold settings and preserving transient shock features, the method significantly enhances both detection accuracy and reliability.

The training data set was constructed using FF events from the CfA shock catalog (1995–2019), while testing used events from the combined CfA and Helsinki catalogs during 2020–2024. In the testing phase, the model identified 169 FF events, of which 83 matched cataloged entries. Among the remaining 86 uncatalogued events, 67 were validated as genuine shocks through Rankine-Hugoniot relations and fast Mach number criteria, confirming their physical authenticity. These results demonstrate the model's ability not only to reproduce existing catalog detections but also to uncover new shock events previously overlooked. As an independent consistency check, we compared our newly identified shocks with the Oliveira list (Oliveira, 2023), which reports the minimum SMR index (Newell & Gjerloev, 2012) within 2 hours after each shock impact. We find that 11 of our newly identified shocks are also listed in Oliveira list. Among these 11 events, three exhibit negative minimum SMR values (min SMR < 0), including one case with a pronounced response (2 May 2024; min SMR ≈ -70 nT). This overlap indicates that our newly identified set complements existing shock catalogs by adding events with measurable geomagnetic activity and by improving catalog completeness, thereby enabling more robust statistical and physical investigations of shock properties and their potential space weather impacts.

Table 5
The 20 Fast Forward Events Undetected by This Method

No.	Shock time (UT)	Catalog	Likely contributing factor
1	2021-11-03 19:35:00	CfA and Helsinki	Lack of data
2	2022-01-18 22:58:00	CfA and Helsinki	No obvious jump
3	2022-03-10 16:11:34	Helsinki	Lack of data
4	2022-05-01 20:18:12	Helsinki	No obvious jump
5	2022-08-17 02:16:23	Helsinki	Lack of data
6	2023-02-14 00:52:43	Helsinki	Lack of data
7	2023-04-09 22:18:47	Helsinki	Lack of data
8	2023-05-07 07:37:19	Helsinki	Missed
9	2023-05-11 21:33:43	Helsinki	No obvious jump
10	2023-08-07 10:48:58	Helsinki	No obvious jump
11	2023-11-05 08:27:20	Helsinki	Lack of data
12	2023-11-25 07:58:43	Helsinki	Lack of data
13	2023-11-30 23:26:00	CfA and Helsinki	Lack of data
14	2023-12-11 11:31:58	Helsinki	Lack of data
15	2023-12-13 21:28:00	CfA and Helsinki	Lack of data
16	2023-12-17 06:54:56	Helsinki	Lack of data
17	2024-01-03 14:19:00	CfA and Helsinki	No obvious jump
18	2024-02-08 19:16:44	Helsinki	No obvious jump
19	2024-02-11 01:23:51	Helsinki	Lack of data
20	2024-02-14 22:16:00	Helsinki	No obvious jump

Performance evaluations reveal substantial improvements over earlier methods. After incorporating Rankine–Hugoniot verification, the model achieved a TS of 0.7438, a recall of 81.62%, an F1-score of 85.31%, and a FAR of only 10.65%. These metrics indicate both high detection capability and robustness against false positives, establishing this approach as a superior tool compared to previous threshold-based schemes.

A key limitation arises when data gaps (NaNs) occur within or near the shock ramp. During training, we excluded samples containing NaNs to ensure stable optimization, because missing values can lead to ill-conditioned gradients and unstable weight updates. As a result, the MLP is trained only on complete, finite valued inputs and is not designed to infer shock signatures across missing segments. In practice, when the most pronounced discontinuity is partially or fully missing and when the required upstream/downstream intervals are affected, reliable identification can be inherently difficult or impossible due to the lack of measurements. Moreover, samples affected by NaNs are assigned to an additional "invalid" label, which can interrupt FF label sequences and reduce the effectiveness of the subsequent ramp identification post-processing. In contrast, traditional threshold-based methods rely on moving-average windows to calculate upstream and downstream parameters. This approach allows them to handle data gaps at shock ramps, but at the cost of smoothing out intrinsic shock features.

Overall, this study highlights the promise of ML in advancing operational space weather forecasting. Automated and accurate shock identification not only enhances capability for space weather forecast but also expands the statistical foundation for investigating the physical properties and heliospheric dynamics of interplanetary shocks. Future extensions may include

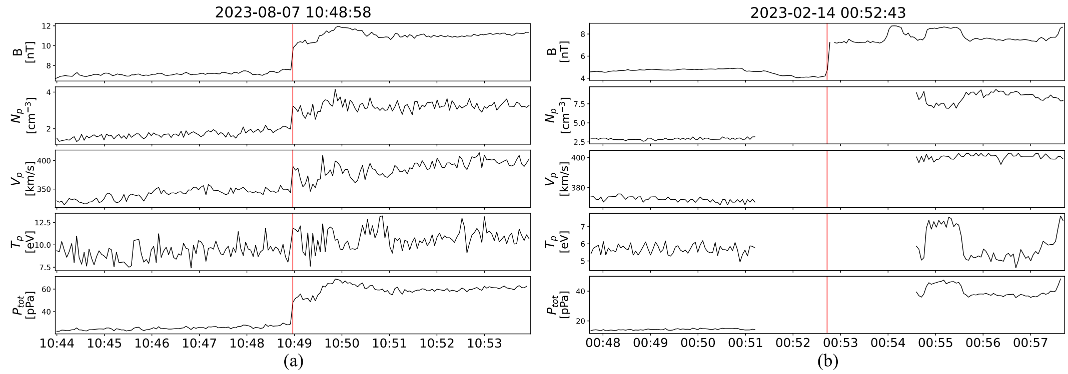

Figure 7. Examples of undetected fast forward (FFs). (a) An undetected FF without obvious heating. (b) An undetected FF due to data gap at the shock ramp. The red vertical lines indicate the FF shock ramps obtained from the combined catalog.

broadening the framework to other types of shocks, integrating data from additional spacecraft (e.g., Parker Solar Probe, Solar Orbiter, Aditya-L1, and SWFO), and incorporating physics-informed constraints to improve both generalization and interpretability.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

The spacecraft data that support the findings of this study are openly available in CDAWeb at https://cdaweb.gsfc.nasa.gov/. The machine-learning algorithm, the training and validation data sets, and the list of novel shocks missing from the combined catalog are publicly archived in the Science Data Bank (Tan et al., 2026). All training hyperparameters are specified in the publicly archived, executable code package.

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